

# Ester Functionalization of Polypropylene via Controlled Decomposition of Benzoyl Peroxide during Solid-State Shear Pulverization

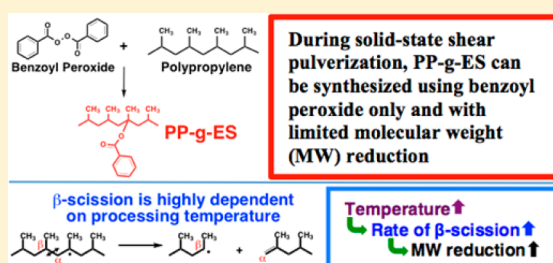
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## S Supporting Information

**ABSTRACT:** Polypropylene (PP) is a nonpolar polyolefin that is sometimes functionalized with polar molecules to allow for reactive compatibilization and improve interfacial adhesion. Functionalized PP is conventionally synthesized by melt extrusion at elevated temperature ( $T$ ) ( $\geq \sim 180$  °C) with a radical initiator and polar monomer, e.g., maleic anhydride. A drawback to high  $T$  functionalization is that  $\beta$ -scission, which leads to cleavage of C–C backbone bonds, is significant and results in major molecular weight (MW) reduction and property degradation. We present a novel method of functionalizing PP using benzoyl peroxide (BPO) alone by a process called solid-state shear pulverization (SSSP), resulting in ester functional groups (benzoates) grafted at high yield onto PP. Ester functionalized PP (PP-g-ES) is synthesized with limited MW reduction because SSSP is done at sufficiently low  $T$  to suppress  $\beta$ -scission. Characterization before and after grafting at 0.18 mol % (0.46 wt %) graft level indicates that functionalization (and subsequent purification) is accompanied by only one chain scission event per 12 400 PP repeat units, resulting in 17% and 36% reductions in number-average MW and weight-average MW, respectively. Benzoate grafting levels are tuned from 0.08 to 0.41 mol % (0.22 to 1.14 wt %) by varying the BPO feed level. In addition to limited MW reduction, PP-g-ES exhibits modified interfacial properties, the ability to undergo transesterification reactions consistent with reactive compatibilization, and little to no loss of physical and mechanical properties relative to neat PP.



## INTRODUCTION

Polyolefins are nonpolar by nature which leads to excellent chemical resistance but poor compatibility with polar molecules. When functionalized with polar moieties<sup>1–15</sup> such as hydroxyl groups,<sup>2,3</sup> itaconates,<sup>4</sup> methacrylate esters,<sup>4,5</sup> and maleic anhydride,<sup>6–12</sup> the modified polyolefins are useful as compatibilizers in blends,<sup>16–18</sup> composites,<sup>19,20</sup> and nanocomposites.<sup>21–23</sup> Commercial, functionalized polyolefins are based on covalent addition of functional monomers to polyolefins in the presence of a radical initiator; the major categories of this type of functionalization are discussed in detail in a recent publication.<sup>12</sup> The most common method used for functionalization is melt extrusion; under the high temperature ( $T$ ) conditions utilized, functionalization is usually accompanied by undesirable side effects, including chain scission, branching, and cross-linking.<sup>24–28</sup> For this reason, there is ongoing interest to develop improved, commercially scalable methods for postpolymerization functionalization of polyolefins in the presence of both a radical initiator, such as an organic peroxide, and a functional monomer.<sup>12</sup>

An alternative method for direct functionalization of polyolefins in the absence of functional monomer was described in a 2005 review by Boen and Hillmyer,<sup>29</sup> who stated, “An interesting approach that is related but could prove

to be more efficient is the direct functionalization with organic peroxides alone.” Attempts to achieve direct functionalization have been limited to the use of asymmetric peroxy derivatives (e.g., acrylic peroxides, peroxyesters, and peroxyketals) that possess an extra functional group (e.g., ester, ketone, carboxylic acid) in addition to the peroxidic O–O bond.<sup>30–37</sup> Following the seminal work by DeNicola et al.<sup>30</sup> in 1995 on direct functionalization of polyolefins using unsaturated peroxides and concluding in 2005, several studies were published regarding polyolefin functionalization with peroxy derivatives.<sup>31–37</sup> These studies mostly relied on thermal decomposition of a peroxy derivative to functionalize a polyolefin at or above the melt  $T$  of the polyolefin.

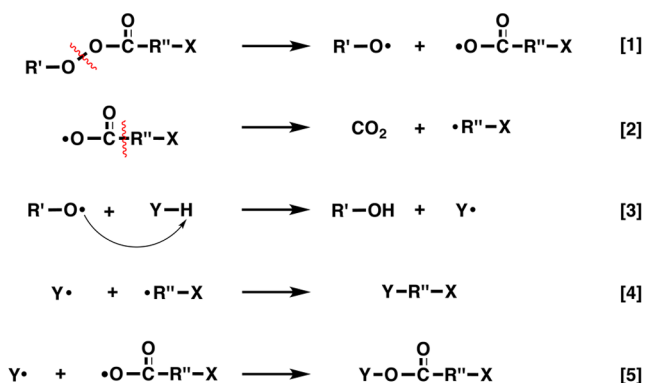
The generalized mechanism for direct functionalization with a peroxyester (see Scheme 1) begins with the dissociation of an asymmetric peroxyester ( $R'O-OCOR''-X$ ) into two radicals,  $R'O\cdot$  and  $\cdot OCOR''-X$ , with  $R'O\cdot$ , an alkoxy radical, being the more reactive of the two radicals formed (Scheme 1, step 1). The asymmetric peroxide is chosen so that it decomposes to form a less reactive radical,  $\cdot OCOR''-X$ , with  $X$  being a

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**Scheme 1. Generalized Mechanism for Direct Functionalization of a Polyolefin with a Peroxy Derivative**



**X = hydrogen or functional groups such as carboxylic acids, esters, or ketones**

**Y-H = polyolefin, e.g., polyethylene, polypropylene**

carbonyl functional group (e.g., carboxylic acid, ester, or ketone). During processing,  $\cdot\text{OCOR}''-\text{X}$  undergoes decarboxylation to produce  $\cdot\text{R}''-\text{X}$  (Scheme 1, step 2). The alkoxy radical,  $\text{R}'\text{-O}\cdot$ , can participate in chain transfer with the polyolefin ( $\text{Y}-\text{H}$ ) to produce a macroradical (Scheme 1, step 3). The polyolefin macroradical ( $\text{Y}\cdot$ ) then terminates with  $\cdot\text{R}''-\text{X}$  to form a functionalized polyolefin. Acrylic peroxides and peroxyketals undergo slightly more complicated mechanisms that include peroxide decomposition induced by polyolefin macroradicals and rearrangements of radicals resulting in polyolefins functionalized with esters, carboxylic acids, or epoxides.<sup>31,32,34,36</sup>

In the late 1990s, Assoun et al.<sup>37</sup> studied polypropylene (PP) carboxylation with peroxy derivatives by melt extrusion at 180 °C in an inert environment. They investigated the effect of the chemical structure of peroxy derivatives on grafting degree,  $\beta$ -scission, and branching.  $\beta$ -scission is highly  $T$  dependent and occurs as a result of rearrangement of a PP macroradical into two chains: one with a terminal double bond and the other with a terminal radical.<sup>24</sup> They determined that even though a highly reactive  $\text{R}'\text{-O}\cdot$  radical (Scheme 1) could improve grafting efficiency, up to ~17%, it resulted in major molecular weight (MW) reduction as a result of  $\beta$ -scission. They also found that the presence of a double bond in the  $\cdot\text{OCOR}''-\text{X}$  radical could lead to branch and/or cross-link formation.

From 2000 to 2005, Maillard and co-workers<sup>31–36</sup> published extensively on direct functionalization of polyethylene (PE) and PP using asymmetric peroxy derivatives, e.g., acrylic peroxides, peroxyesters, peroxyketals, and unsaturated peroxides. Using a 2.5 h batch process in an inert environment at 160 °C and atmospheric pressure, they achieved ester functionalization of PE with peroxyesters and peroxyketals.<sup>31–33</sup> In addition to functionalizing PE, they observed varying degrees of cross-linking depending on the nature of the radicals formed after peroxide decomposition. They also showed that functionalization of isotactic and atactic PP was possible using peroxyesters or peroxyketals and a 160 °C batch process.<sup>34,35</sup> Although they expected greater functionalization of PP relative to PE because the C–H bond dissociation energies are lower for methane hydrogens than for methyl or methylene hydrogens, they found that hydrogen abstraction (needed for functionalization) “hardly occurs”<sup>34</sup> with PP and inferred that PP methyl groups hinder hydrogen abstraction by 1,1-dimethylethoxy radicals.<sup>34</sup>

Maillard and co-workers<sup>31–36</sup> and Assoun et al.<sup>37</sup> acknowledged the possibility of achieving direct functionalization with an acyloxy radical,  $\cdot\text{OCOR}''-\text{X}$ , i.e., without undergoing decarboxylation to form an alkyl radical,  $\cdot\text{R}''-\text{X}$ . Indeed, this type of termination reaction is possible as long as the alkyl radical is more stable than its acyloxy counterpart.<sup>38,39</sup> The tendency of an acyloxy radical to undergo decarboxylation depends on  $T$ , pressure, and the stability of the alkyl radical that is formed.<sup>38,39</sup> For most diacyl peroxides, decarboxylation happens almost simultaneously with the cleavage of the O–O bond at all  $T$ 's; this is not so for diacetyl peroxide or benzoyl peroxide (BPO), which both form unstable alkyl radicals (methyl and phenyl, respectively) after decarboxylation.<sup>38</sup> Additionally, the presence of a double bond stabilizes an acyloxy radical and reduces the rate of decarboxylation.<sup>37</sup> It is under one of these conditions (i.e., the formation of radicals from unsaturated asymmetric peroxides) that Assoun et al.<sup>37</sup> obtained data that suggested that small levels of direct functionalization of PP with esters from acyloxy radicals ( $\cdot\text{OCOR}''-\text{X}$  in Scheme 1) had been achieved. Chateaufort et al.<sup>40</sup> demonstrated that for benzoyloxy radicals formed from BPO decomposition the rate constant for decarboxylation depends strongly on  $T$ , increasing by ~200-fold as  $T$  increases from 25 to 200 °C. Thus, a potential approach for increasing the grafting degree of acyloxy radicals (in particular, benzoyloxy radicals) during polyolefin functionalization is to decrease the processing  $T$ . By so doing, the extent of  $\beta$ -scission, and thus MW reduction, can be decreased as well.

Here, we show for the first time that a symmetric organic peroxide, BPO, can be used in the absence of a polar monomer to graft polar functional groups onto PP. In particular, we demonstrate that solid-state shear pulverization (SSSP), a low  $T$ , continuous process with potential for high throughput, provides a tunable route to synthesize ester functionalized PP (PP-g-ES). This process uses a twin-screw melt extruder modified with a cooling system to maintain the polymer in the solid state.<sup>12,41–50</sup> Solid-state shear pulverization is accompanied by high shear stresses and compressional forces that cause repeated fragmentation and fusion of material; conditions are tuned by feed rate, screw speed, screw design, and  $T$ .<sup>42</sup> In addition to being solventless, SSSP is industrially scalable. (Polyolefins have been processed by SSSP with a commercial-scale apparatus at rates exceeding 150 kg/h.) Compatibilization and intimate mixing of immiscible blends<sup>43–46</sup> and effective dispersion and exfoliation in composites and nanocomposites can result from SSSP.<sup>47–51</sup> Very recently, we also showed that SSSP can be used to synthesize maleic anhydride grafted PP with strongly suppressed MW reduction.<sup>12</sup> In the present study, by using SSSP for postpolymerization synthesis of PP-g-ES, we present a platform of low- $T$  chemistries that is unattainable with high- $T$  melt processing. The functionalization results in benzoate side groups that can modify interfacial interactions with polar molecules and result in a reactive PP of the type useful in reactive compatibilization.

## EXPERIMENTAL SECTION

**Materials.** Polypropylene (Total Petrochemicals; MFI = 1.5 g/10 min; ASTM standard D-1238 at 230 °C/2160 g load; reported by the supplier) was used as received. Benzoyl peroxide was used as received (SigmaAldrich). Xylene, octadecyl acrylate (ODA), and 1-pyreneacetic acid (Pyr-AA) were used for characterizing grafting degrees and reactivity of the PP-g-ES and were used as received (SigmaAldrich). A

**Table 1. Variables for Sample Composition, Processing Method, and Characterization of Benzoate Grafting Degrees, Crystallinity, and Tensile Properties for Neat PP and PP-g-ES Samples Made via SSSP**

| sample                        | amount of BPO Added (wt %) | processing method | grafting degree <sup>a</sup> |                      | grafting yield (%) | crystallinity (%) | Young's modulus (MPa) | yield strength (MPa) |
|-------------------------------|----------------------------|-------------------|------------------------------|----------------------|--------------------|-------------------|-----------------------|----------------------|
|                               |                            |                   | (wt %)                       | (mol %) <sup>b</sup> |                    |                   |                       |                      |
| neat PP pellets (as received) |                            |                   |                              |                      |                    | 45                | 1250 ± 80             | 36 ± 1               |
| neat PP (after SSSP)          |                            | SSSP              |                              |                      |                    | 44                | 1170 ± 60             | 35 ± 1               |
| PP-g-ES/1                     | 0.5                        | SSSP              | 0.22                         | 0.08                 | 87                 | 41                | 1270 ± 30             | 36 ± 1               |
| PP-g-ES/2                     | 1.0                        | SSSP              | 0.33                         | 0.12                 | 65                 | 42                | 1190 ± 30             | 34 ± 1               |
| PP-g-ES/3                     | 1.5                        | SSSP              | 0.46                         | 0.18                 | 61                 | 41                | 1130 ± 30             | 32 ± 1               |
| PP-g-ES/4                     | 3.0                        | SSSP              | 0.81                         | 0.30                 | 52                 | 42                | 1190 ± 60             | 32 ± 2               |
| PP-g-ES/5                     | 6.0                        | SSSP              | 1.14                         | 0.41                 | 36                 | 40                | 1080 ± 40             | 29 ± 1               |
| PP-g-ES/4MM                   | 3.0                        | melt mixing       |                              |                      |                    |                   |                       |                      |

<sup>a</sup>Grafting degree can also be expressed in  $\mu\text{equiv}$  (i.e.,  $\mu\text{mol}$  of benzoates in 1 g of PP-g-ES) by simple stoichiometric calculations (e.g., 0.18 wt% is equivalent to 0.22  $\mu\text{equiv}$  benzoate grafted onto PP). <sup>b</sup>Functionalization in mol % represents the number of moles of benzoates per mole of repeat units multiplied by 100%.

phenolic antioxidant, Songnox 6260 (Songwon), was used as received in samples made for rheological characterization.

**Synthesis of PP-g-ES with SSSP.** The PP and BPO were pulverized using a relatively harsh screw design<sup>42</sup> at 200 rpm screw speed and 100 g/h feed rate. The pulverizer was a pilot-plant/research scale Berstoff twin-screw extruder (screw diameter = 25 mm, length/diameter = 26.5) modified with a cooling system (a Budzar Industries WC-3 chiller operating at  $-6\text{ }^{\circ}\text{C}$ ); the same apparatus was used in previous SSSP studies.<sup>12,41–50</sup> (It should be noted that the  $T$  of the solid-state polymer in the pulverizer may, at various locations, exceed room temperature by several tens of degrees and be warm to the touch upon exiting the pulverizer.) Samples of PP-g-ES were prepared by SSSP using 0.5–6.0 wt % BPO in the feed. A PP-g-ES sample prepared with 3 wt % BPO was also prepared by melt processing for 10 min at  $200\text{ }^{\circ}\text{C}$  with an Atlas Electronic Devices MiniMAX molder (cup-and-rotor mixer) at maximum rotor speed and with three steel balls in the cup to provide chaotic mixing.<sup>52</sup> Table 1 shows sample composition and process methods.

**Purification of PP-g-ES.** To ensure high purity and the absence of contaminants, PP-g-ES samples employed for quantification of carbonyl grafting, contact angle measurements, and demonstration of reactivity were purified to remove unreacted BPO by dissolution in boiling xylene followed by precipitation with methanol. The samples were dried under vacuum at  $70\text{ }^{\circ}\text{C}$  for 24 h. The PP-g-ES samples for rheology were purified using a variety of methods that combined washing with acetone and annealing under vacuum; samples were annealed either at room  $T$  for 96 h or at  $50\text{ }^{\circ}\text{C}$  for 48 h. For each annealing condition, two sets of samples were studied: one set was washed with acetone before annealing, and the other was not washed before annealing. For the washing step, 100 g of PP-g-ES was stirred in 300 mL of acetone for 3 h at room  $T$ ; the PP-g-ES was then removed by filtration. The PP-g-ES samples used in high- $T$  gel permeation chromatography (high- $T$  GPC) and physical and mechanical property characterization were purified by washing with acetone and annealing at  $50\text{ }^{\circ}\text{C}$  for 48 h.

**Quantification of Carbonyl Grafting.** In order to create a calibration curve, ODA/PP blends were prepared by melt processing at  $200\text{ }^{\circ}\text{C}$  in a MiniMAX molder for 10 min and at maximum rotor speed with three steel balls in the cup in order to provide chaotic mixing.<sup>52</sup> Blend products were compression molded into thin films ( $\sim 0.3\text{ mm}$  thick) for Fourier transform infrared (FTIR) spectroscopy using a PHI hot press coupled with a PHI cold press. Octadecyl acetate was chosen because of its structural similarity to PP-g-ES. For each blend, three sets of FTIR data were collected with 64 scans and  $4\text{ cm}^{-1}$  resolution. Purified PP-g-ES samples were compression molded into thin films ( $\sim 0.3\text{ mm}$  thick) and tested under the same conditions as ODA/PP blends.

**Rheological Measurements.** Neat PP and PP-g-ES samples were tested with 0.5 wt % Songnox 6260 added to each sample to prevent

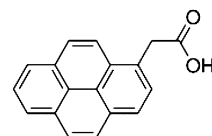
thermal degradation. Samples were compression molded into disks devoid of bubbles using PHI presses. Small-amplitude oscillatory shear data were collected at  $180\text{ }^{\circ}\text{C}$ , with 2% strain over a frequency range of 0.01–100 rad/s (measuring from high to low frequency), using a strain-controlled Rheometrics Scientific ARES rheometer equipped with 25 mm parallel plates.

**Physical and Mechanical Properties.** Thermal analysis employed a Mettler Toledo differential scanning calorimeter (DSC 822e). Samples were heated at  $40\text{ }^{\circ}\text{C}/\text{min}$  to  $200\text{ }^{\circ}\text{C}$ , held at  $200\text{ }^{\circ}\text{C}$  for 5 min, cooled at  $40\text{ }^{\circ}\text{C}/\text{min}$  to  $40\text{ }^{\circ}\text{C}$ , held at  $40\text{ }^{\circ}\text{C}$  for 3 min, heated at  $10\text{ }^{\circ}\text{C}/\text{min}$  to  $200\text{ }^{\circ}\text{C}$ , held at  $200\text{ }^{\circ}\text{C}$  for 5 min, and cooled at  $10\text{ }^{\circ}\text{C}/\text{min}$  to  $40\text{ }^{\circ}\text{C}$ . The crystallinity was determined from the final cooling step.

Films with  $\sim 0.7\text{ mm}$  thickness were prepared by pressing in a PHI hot press at  $200\text{ }^{\circ}\text{C}$  for 5 min and then rapidly cooling in a PHI cold press for 15 min. Tensile specimens were prepared according to ASTM D1708; dumbbell-shaped specimens were cut from films using a Dewes-Gumbs die. An MTS Sintech 20/G (100 kN load cell; crosshead speed =  $5\text{ cm}/\text{min}$ ) was used to obtain Young's modulus and yield strength values at room temperature.

**Contact Angle Measurement.** Neat PP and PP-g-ES samples were compression molded into thin films ( $\sim 0.3\text{ mm}$  thick) with smooth surfaces using a PHI hot press at  $200\text{ }^{\circ}\text{C}$  for 5 min and then rapidly cooling in a PHI cold press for 10 min. The static contact angle of a droplet of deionized water ( $5\text{ }\mu\text{L}$ ) on the surface of each disk was determined using a KRUS drop shape analysis system, DSA 100; results for each sample were averaged over 20 measurements. Reported values of contact angle measurements have  $\pm 1^{\circ}$  experimental errors.

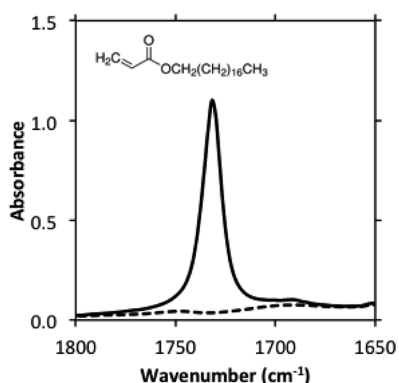
**Demonstration of Reactivity of PP-g-ES with Pyr-AA.** 10 g/L PP-g-ES samples were dissolved in 0.30 g/L solutions of Pyr-AA in xylene. Solutions were held at  $130\text{ }^{\circ}\text{C}$  for 4 h, after which PP-g-ES was precipitated with methanol. To remove unreacted Pyr-AA, samples were purified six times by dissolution in boiling xylene and precipitation in methanol. Pyrene label fluorescence was measured with a Photon Technology International fluorimeter ( $\lambda_{\text{exc}} = 344\text{ nm}$ ).



1-pyreneacetic acid (Pyr-AA)

## RESULTS AND DISCUSSION

### Quantitative Characterization of Ester Functionalization Levels on PP-g-ES: FTIR Spectroscopy. Figure 1



**Figure 1.** FTIR spectra of an ODA/PP blend containing 155  $\mu$ equiv of ODA, i.e., 5 wt % ODA in ODA/PP blend (solid curve) and neat PP (dashed curve). The inserted molecular structure is that of octadecyl acrylate (ODA).

compares FTIR spectra of neat, as-received PP and a sample containing 5.0 wt % ODA in an ODA/PP blend. The peak at  $1732\text{ cm}^{-1}$  is present only for the ODA/PP blend and is associated with the ODA ester functional group.<sup>53</sup> The inset in Figure 1 is the chemical structure of ODA. A peak at  $841\text{ cm}^{-1}$ , specific to PP and absent for BPO, is used for normalization of each sample spectrum. For spectral analyses, the data between  $1800$  and  $1600\text{ cm}^{-1}$  and between  $930$  and  $740\text{ cm}^{-1}$  were deconvoluted into component peaks using a Lorentzian function. This yielded accurate peak intensities while accounting for peak overlaps and inconsistent baselines between spectra. A calibration curve based on ODA was determined using

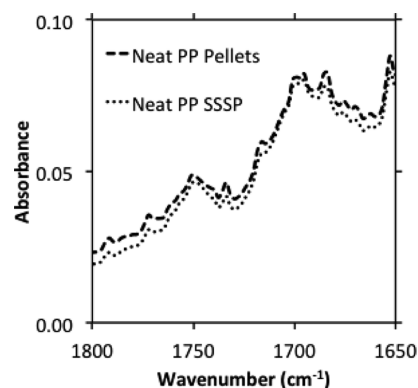
$$[\text{ester}] = 0.25(A_{1732}/A_{841}) \quad (1)$$

where  $[\text{ester}]$  is the grafting degree and measured in mol % (i.e., moles of benzoate per mole of PP repeat unit) and  $A_{1732}$  and  $A_{841}$  are the areas under the peaks at  $1732$  and  $841\text{ cm}^{-1}$ , respectively. The grafting yield was calculated using

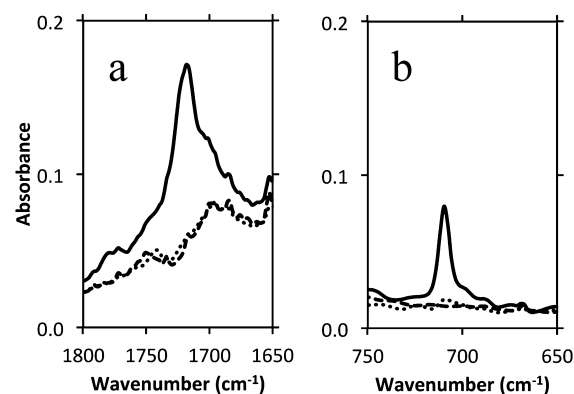
$$\text{grafting yield} = ([\text{ester}]/[\text{BPO}]_0) \times 100\% \quad (2)$$

where  $[\text{ester}]$  is as defined in eq 1 and  $[\text{BPO}]_0$  is the initial amount of BPO added during SSSP; both variables were measured in mol %. (It must be noted that if each of the two benzoyloxy radicals resulting from the dissociation of BPO were to be grafted to PP, then the grafting yield would be 200%. Of course, for this to occur, all PP radicals would have to be generated solely by chain scission accompanying SSSP, without any chain transfer reactions. Chain scission has been shown to accompany SSSP of PP, albeit at quite limited levels, under harsh pulverization conditions and decreasing with decreasing levels of work done on the PP during SSSP.<sup>42,43</sup>) Grafting degrees and yields of PP-g-ES samples were determined using eqs 1 and 2 and are reported in Table 1; grafting degrees are presented in both mol % and wt %.

Figure 2 compares the FTIR spectra for neat, as-received PP, and neat PP after SSSP. Because of the strong similarity between the spectra in the regions of interest ( $1800$ – $1600\text{ cm}^{-1}$ ), we can conclude that no functionalization due to ketonization or carboxylic acid formation (as a result of radical stabilization by atmospheric oxygen<sup>54</sup>) occurs during SSSP processing of neat PP. Figure 3 compares the FTIR spectra of PP-g-ES/4, PP-g-ES/4MM, and neat PP pellets and confirms the presence of a strong absorbance peak for PP-g-ES/4 at  $1720$



**Figure 2.** FTIR spectra of neat PP as received (dashed curve) and neat PP after SSSP (dotted curve).



**Figure 3.** FTIR spectra of PP-g-ES/4 (solid curve), PP-g-ES/4MM (dotted curve), and neat PP as received (dashed curve) in spectral ranges of (a)  $1800$ – $1650$  and (b)  $750$ – $650\text{ cm}^{-1}$ .

$\text{cm}^{-1}$  and the absence of this absorption for the melt processed sample (see Figure 3a). As explained above, this peak is associated with ester functional groups; the slight shift of this peak to a lower wavenumber (as compared to the ester peak observed for ODA) is expected for aromatic esters.<sup>53</sup> In Figure 3b, we see an absorbance at  $\sim 740\text{ cm}^{-1}$  for PP-g-ES/4. This absorbance is associated with phenyl group attachment to the PP backbone and is absent for PP-g-ES/4MM, which confirms that no significant functionalization of an ester or addition of a phenyl group can be achieved using BPO via high- $T$  melt processing. The absence of functionalization during melt processing can be understood from the  $T$  dependence of BPO dissociation into radicals.<sup>40,55</sup> A detailed explanation for this is given in the next subsection.

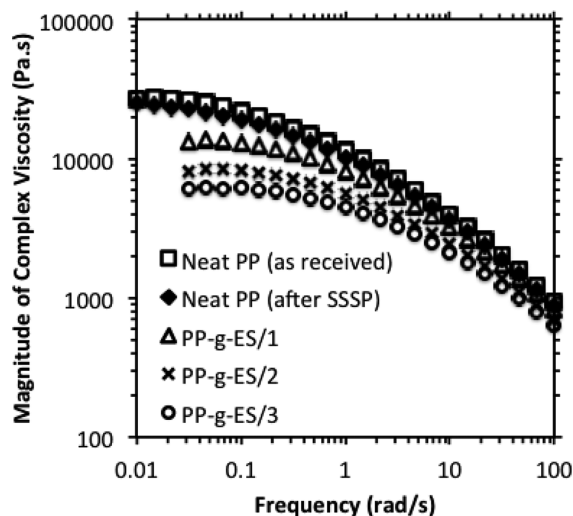
As compared to grafting yields reported for PP-g-ES synthesized with peroxyesters<sup>34,37</sup> and unsaturated peroxides<sup>35</sup> at  $160$ – $180\text{ }^{\circ}\text{C}$  and in an inert environment, the 36–87% grafting yields (corresponding to 0.41–0.18 mol % grafting degrees, respectively) obtained during PP-g-ES synthesis via SSSP in an air environment are significantly higher. Assoun et al.<sup>37</sup> reported  $\sim 17\%$  (and 0.02 mol %) as the largest grafting yield for a series of peroxyesters used for PP functionalization by melt extrusion at  $180\text{ }^{\circ}\text{C}$  and in an  $\text{N}_2$  environment. Saule et al.<sup>34</sup> reported a grafting yield of 5% for isotactic PP grafted in a 2.5 h batch process carried out in an inert environment at  $160\text{ }^{\circ}\text{C}$  using a peroxyester. For a series of unsaturated peroxides, which were expected to significantly improve grafting yields in PP, Saule et al.<sup>35</sup> reported a maximum grafting yield of  $\sim 40\%$ .<sup>56</sup> It is evident from these comparisons that PP-g-ES synthesis



via melt processing because peroxy derivatives are chosen so that their radicals are very good at abstracting hydrogen atoms from the PP backbone to form PP macroradicals. At melt-state  $T$ , PP macroradicals have a high propensity to undergo  $\beta$ -scission. Thus, postpolymerization functionalization of PP by melt processing results in dramatic MW reduction. For example, for PP-g-ES samples made by melt processing with asymmetric peroxy derivatives, Saule et al.<sup>35</sup> reported weight-average MW ( $M_w$ ) reductions of 53–70% with grafting yields of 30–40%,<sup>56</sup> and Assoun et al.<sup>37</sup> reported a 24% reduction in  $M_w$  at only 0.02 mol % grafting level and  $\sim$ 17% grafting yield. (Grafting yields have a maximum of 100% when grafting is done with asymmetric peroxy molecules.) Here, using rheology, we evaluate the change in PP  $M_w$  caused by benzoate functionalization and consider the effect of the method used to purify the PP-g-ES of unreacted BPO and its byproducts.

During SSSP, shear stresses and compressional forces cause repeated fragmentation and fusion of the material in the solid state.<sup>42</sup> Thus, we expect that unreacted BPO will be located not only on the surface of PP powder particles but also trapped within the particles. The products of SSSP processing of PP with BPO, resulting in 0.08–0.41 mol % ester functionalization and grafting yields as high as 87%, were purified of unreacted BPO and its byproducts in a number of ways prior to characterization by rheology: no purification; annealing samples under vacuum at room temperature for 96 h to purify by sublimation;<sup>64–67</sup> washing with acetone to remove surface BPO and then annealing under vacuum at room temperature for 96 h; annealing samples under vacuum at 50 °C for 48 h to purify by decomposition and/or sublimation;<sup>55,64–67</sup> and washing with acetone to remove surface BPO and then annealing under vacuum at 50 °C for 48 h.

Figure 4 compares the magnitudes of complex viscosity ( $|\eta^*|$ ) of PP-g-ES/1, PP-g-ES/2, and PP-g-ES/3 that were



**Figure 4.** Magnitude of complex viscosity as a function of frequency for as-received neat PP pellet ( $\square$ ), neat PP pulverized ( $\blacklozenge$ ), PP-g-ES/1 ( $\triangle$ ), PP-g-ES/2 ( $\times$ ), and PP-g-ES/3 ( $\circ$ ). Data were collected at 180 °C.

washed with acetone and annealed under vacuum at 50 °C for 48 h to those of control samples. The three PP-g-ES samples as well as the control samples exhibited  $|\eta^*|$  values that were independent of frequency at low frequency. With such behavior and application of the Cox–Merz rule,<sup>68</sup> which relates linear

viscoelastic properties to steady shear viscosity, and the Cross model,<sup>69</sup> it is possible to determine the zero shear rate viscosity ( $\eta_0$ ) from  $|\eta^*|$  data. Assuming that  $\eta_0$  scales with  $M_w$  to the 3.4 power,<sup>70–73</sup> we can then use rheology to characterize the effect of SSSP processing, functionalization, and post-SSSP purification method on  $M_w$ .

Before discussing such characterization of  $M_w$  in detail, it is important to note that some samples did not exhibit the presence of a  $\eta_0$  regime in the rheology data. Instead, even at the lowest frequency,  $|\eta^*|$  was increasing with decreasing frequency. (Selected examples of this behavior are shown in the Supporting Information.) This behavior is believed to be associated with long-chain branching. Examples fall into two classes: (1) PP-g-ES/2 and PP-g-ES/3 that were not subjected to any purification prior to rheological testing and (2) PP-g-ES/4 and PP-g-ES/5 that were subjected to all levels of purification, which indicates that these samples, with the highest ester functionalization levels, were branched coming out of the SSSP apparatus. This branching behavior will be addressed in more detail in future work.

Neat PP after SSSP had  $\eta_0$  that was 10% lower than that of neat PP pellets (i.e.,  $\eta_0 = 28,500$  Pa·s for neat PP after SSSP), indicating a 3% reduction in  $M_w$ . Thus, in the absence of BPO, SSSP causes nearly negligible MW reduction of neat PP. For PP-g-ES/1 (0.08 mol % graft level and 87% grafting efficiency) that was characterized directly as the output from the pulverizer without any purification step to remove unreacted BPO,  $\eta_0$  was reduced by  $\sim$ 61% relative to neat PP pellets, indicating a 24% reduction in  $M_w$ . This  $M_w$  reduction is similar to the 24% reduction measured by Assoun et al.<sup>37</sup> for a PP-g-ES sample made by melt processing and asymmetric peroxy molecules which had only 0.02 mol % functionalization, one-fourth that of our PP-g-ES/1 sample. When the same PP-g-ES/1 sample was subjected to various purification steps, the apparent reductions in  $M_w$  from zero shear rate viscosity data were 11–12% (unwashed or washed with acetone, followed by room temperature annealing under vacuum for 96 h) and 20–21% (unwashed or washed with acetone, followed by annealing at 50 °C under vacuum for 48 h). The results indicate that at low functionalization levels only very small amounts of  $\beta$ -scission leading to MW reduction occur during SSSP; this is also true even during post-SSSP annealing or processing under conditions that would lead to decomposition of unreacted BPO into radicals (i.e., annealing at 50 °C).

At moderate functionalization, PP-g-ES/2 and PP-g-ES/3 with 0.12–0.18 mol % ester, which require higher BPO levels in the feed, linear chain behavior is evident from rheology for all purification conditions except no purification. This implies that the PP-g-ES output from the pulverizer is linear in nature and only becomes branched if substantial levels of covalently attached benzoates and undecomposed BPO are present in the sample at the time it is subjected to melt processing. Based on  $\eta_0$  values,  $M_w$  was reduced relative to neat PP by 23–25% and 27–28% in PP-g-ES/2 and PP-g-ES/3 for samples (washed with acetone or unwashed and) annealed at room temperature under vacuum for 96 h and by 30% and 36% in PP-g-ES/2 and PP-g-ES/3 for samples (washed with acetone or unwashed and) annealed at 50 °C under vacuum for 48 h. While these values of  $M_w$  reduction are not negligible, they are far below the 53–70% reductions reported by Saule et al.,<sup>35</sup> who did postpolymerization ester functionalization of PP by melt processing with asymmetric peroxy molecules.

Table 2. MW Characterization before and after Functionalization via SSSP Using Oscillatory Shear Rheology and High-*T* GPC

| sample                        | grafting deg<br>(mol %) | rheology        |                                     | high- <i>T</i> GPC |                        |                  |                        |       | ester groups per<br>chain |
|-------------------------------|-------------------------|-----------------|-------------------------------------|--------------------|------------------------|------------------|------------------------|-------|---------------------------|
|                               |                         | $\eta_0$ (Pa·s) | $M_w$ reduction <sup>a</sup><br>(%) | $M_n$<br>(g/mol)   | $M_n$ reduction<br>(%) | $M_w$<br>(g/mol) | $M_w$ reduction<br>(%) | $z_c$ |                           |
| neat PP pellets (as-received) |                         | 31 800          |                                     | 115 000            |                        | 585 300          |                        |       |                           |
| PP-g-ES/1                     | 0.08                    | 14 800          | 20                                  | 104 900            | 9                      | 439 800          | 25                     | 0.10  | 2                         |
| PP-g-ES/2                     | 0.12                    | 9 900           | 29                                  | 104 400            | 9                      | 422 800          | 28                     | 0.10  | 3                         |
| PP-g-ES/3                     | 0.18                    | 7 100           | 36                                  | 95 400             | 17                     | 376 000          | 36                     | 0.21  | 4                         |

<sup>a</sup>Percent reduction in  $M_w$  relative to neat PP pellets (as received) was calculated using the assumption that  $\eta_0$  scales with  $M_w$  to the 3.4 power.

**High-*T* GPC Characterization of PP-g-ES and Chain Scission Events per PP Repeat Unit.** High-*T* GPC data were obtained for neat PP pellets and the linear PP-g-ES/1, PP-g-ES/2, and PP-g-ES/3.<sup>74</sup> In order to draw a comparison that reflects chain scission that could occur during the purification process, we evaluated those samples that had been washed with acetone and annealed at 50 °C under vacuum for 48 h and which showed larger reductions in  $\eta_0$  than samples that had been annealed at room temperature for 96 h. Samples were dissolved in trichlorobenzene and tested at 145 °C; data from a triple detection method were used in this paper. (It is important to note that data associated with triple detection provided the worst case scenario for percent reductions in MW as compared to data from other detection methods.<sup>75</sup>) Average MW values, as well as percent reduction in number-average MW ( $M_n$ ) and percent reduction in  $M_w$ , for these samples are presented in Table 2. The percent reductions in  $M_w$ , as determined from high-*T* GPC, are in good agreement with those determined from rheology (see Table 2). As expected, the percent reduction in  $M_w$  is greater than the corresponding percent reduction in  $M_n$  for each sample. This is because the probability that a chain undergoes scission increases as the chain length increases, thereby resulting in a greater reduction in  $M_w$  than  $M_n$ .

The average number of scission events per chain can be determined from the  $M_n$  data:<sup>12</sup>

$$z_c = [M_{n,0}/M_{n,f}] - 1 \quad (3)$$

where  $M_{n,0}$  is the initial  $M_n$ , and  $M_{n,f}$  is the final  $M_n$  after scission (and purification). Based on eq 3,  $z_c = 0.10$  scission events per chain for PP-g-ES/1, i.e., for every 100 original PP chains, there were 10 scission events. If we assume that each scission event resulted in the formation of a radical at each chain end,<sup>76</sup> then we can expect a total of 20 benzoates to be grafted onto 110 PP chains. However, on the basis of the 104 900 g/mol  $M_n$  and 0.08 mol % grafting degree for PP-g-ES/1, we determine that ~2 benzoates are grafted per chain. Thus, for 110 chains, PP-g-ES/1 contains ~220 grafted benzoates, a factor of 11 more than the number of chain ends created. This indicates that the vast majority of benzoates are grafted along the PP backbone and not only at chain ends, that little of the benzoate grafting is due to reaction of benzoyloxy radicals with PP radicals formed by chain scission accompanying SSSP, and that the limit for grafting yield during SSSP (as defined by eq 2) is at most slightly above 100%.<sup>76</sup> This also provides further proof that during SSSP PP macroradicals are less likely to undergo  $\beta$ -scission, thus allowing for benzoate functionalization along the PP backbone. Using similar analyses,  $z_c$  and the number of benzoates per chain for PP-g-ES/2 and PP-g-ES/3 were determined and are shown Table 2. For both PP-g-ES/2 and PP-g-ES/3, the number of benzoates grafted onto the PP

chain vastly exceeds what would have been expected if the benzoyloxy radical only terminated with PP macroradicals produced as a result of chain scission accompanying SSSP.

With  $z_c = 0.10$  for PP-g-ES/1 and PP-g-ES/2 and noting that  $M_{n,0} = 115 000$  g/mol for the neat PP starting material, we determine that one scission event happens every 26 100 repeat units. A similar calculation indicates that one scission event happens every 12 400 repeat units in the synthesis (and purification) of PP-g-ES/3. For PP-g-ES syntheses by Saule et al.<sup>35</sup> at 160 °C in inert environments and resulting in 30–40% grafting yields, we calculate that there was one scission event per 1000–2000 repeat units. Thus, PP-g-ES synthesis by SSSP results not only in higher grafting yield (62–87% for PP-g-ES/1, PP-g-ES/2, and PP-g-ES/3) but also in a factor of ~10 lower frequency of scission events as compared to PP-g-ES prepared by Saule et al.<sup>35</sup>

For the PP-g-ES sample with only 0.02 mol % ester functional group synthesized by Assoun et al.<sup>37</sup> by melt processing in an inert environment, we calculate that one scission event occurred for every 11 800 repeat units. In comparison with PP-g-ES/1 (prepared with similar peroxide feed composition of ~0.1 mol %), the PP-g-ES prepared by Assoun et al.<sup>37</sup> showed not only a much lower grafting degree as compared to its SSSP counterpart (0.08 mol % grafting degree and one scission event per 26 100 repeat units) but also a factor of 2 higher frequency of chain scission. Thus, relative to postpolymerization synthesis by melt processing, PP-g-ES synthesis by SSSP results in enhanced grafting degree and yield and in a major reduction in the frequency of chain scission events per repeat unit, which in turn causes MW reduction.

Given our recent success in synthesizing maleic anhydride modified PP (PP-g-MA) via SSSP of PP in the presence of MA and azobis(isobutyronitrile) (AIBN) as radical initiator,<sup>12</sup> it is also important to compare the level of chain scission events that occur during the SSSP syntheses of PP-g-ES and PP-g-MA. For a PP-g-MA sample with 0.21 mol % MA functionalization, high-*T* GPC analysis revealed that one scission event occurred for every 24 500 repeat units,<sup>12</sup> which is approximately one-half the level of chain scission in PP-g-ES/3 (one per 12 400 repeat units) with 0.18 mol % ES functionalization. The increase in the frequency of scission events for PP-g-ES/3 as compared to its PP-g-MA counterpart (both synthesized via SSSP) can be attributed to the fact that BPO radicals participate in chain transfer with PP to a much greater extent than radicals from AIBN (the radical initiator used to synthesize PP-g-MA via SSSP).<sup>77</sup> Nevertheless, it is evident that polar functionalization of PP via SSSP affords us the ability to suppress the frequency of chain scission events (and hence suppress MW reduction) by taking advantage of the low-*T* processing conditions associated with SSSP.

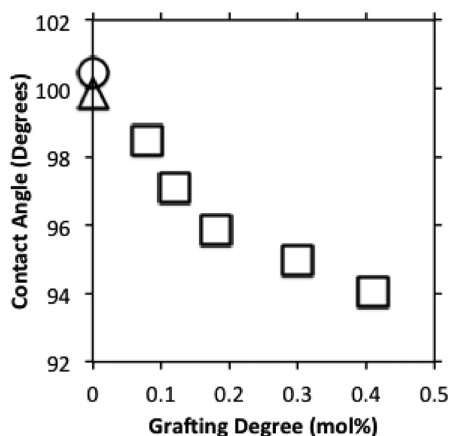
**Physical and Mechanical Properties.** Table 1 shows the percent crystallinity for PP-g-ES samples made by SSSP. Percent crystallinity ( $\chi_{\text{crys}}$ ) was determined using eq 4:

$$\chi_{\text{crys}} = (\Delta H_f / \Delta H_f^\circ) \times 100\% \quad (4)$$

where  $\Delta H_f$  is the sample enthalpy of fusion and  $\Delta H_f^\circ$  is enthalpy of fusion for 100% crystalline PP ( $\Delta H_f^\circ = 207.1 \text{ J/g}^{78}$ ). Within experimental error, we observe that SSSP of neat PP does not affect crystallinity. For PP-g-ES samples, there are only slight decreases in crystallinity relative to neat as-received PP, from 45% to 40–42%. This behavior can be explained by the fact that the bulky benzoate groups grafted onto the PP backbone (and possibly the presence of PP branches) disrupt crystal formation.

Table 1 also compares Young's modulus ( $E$ ) and yield strength ( $\sigma_y$ ) values of neat PP before and after SSSP with those of PP-g-ES samples. Within error, SSSP of neat PP had no effect on  $E$  and  $\sigma_y$ , consistent with the fact that the MW and crystallinity of neat PP before and after SSSP were little changed or identical within error. The grafting of PP with 0.08–0.30 mol % benzoate by SSSP results in no change  $E$  within error from the value for neat as-received PP and, at most, only a  $\sim 10\%$  reduction in  $\sigma_y$ . However, with 0.41 mol % ester functionalization (sample PP-g-ES/5), there is a slightly less than 15% decrease in  $E$  and a 20% decrease in  $\sigma_y$  relative to neat as-received PP, both outside experimental error. These small reductions may be expected as tensile properties of PP are closely linked to MW and crystallinity, both of which decrease slightly with benzoate grafting and are lowest for PP-g-ES/5.<sup>79</sup>

**Interfacial Property Modification: Contact Angle Characterization.** In order to verify the improved wettability and polarity of PP-g-ES samples as compared to neat as-received PP, contact angles were measured for sessile drops of deionized water placed on the surfaces of smooth neat PP and PP-g-ES films (see Figure 5). The contact angle decreases



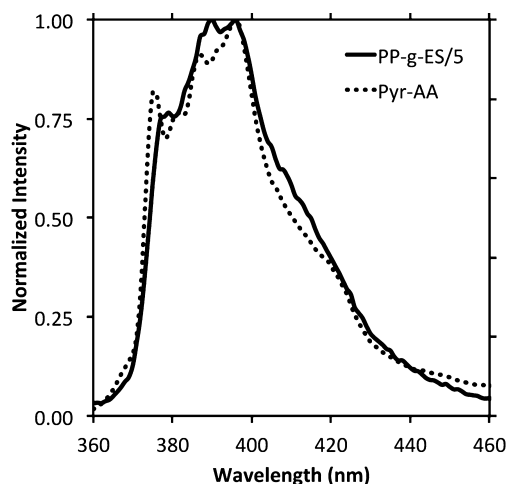
**Figure 5.** Contact angle measurements of neat PP pellets as received (O), neat PP after pulverization (Δ), and PP-g-ES samples synthesized via SSSP with varying degrees of ester functionalization (□).

smoothly with increasing level of ester grafting, which is consistent with the notion that functionalized PP will have improved interfacial adhesion with more polar materials as compared to the parent PP from which it was synthesized.<sup>80,81</sup> In contrast, the contact angle of neat PP after SSSP remains unchanged, within error, as compared to the neat, as-received

PP. Thus, the polarity of neat PP is unaffected by SSSP processing; this is consistent with our observation of the absence of ketonization or carboxylic acid formation from any radical stabilization by atmospheric oxygen.<sup>54</sup> These results provide further confirmation of the ability of SSSP to achieve direct ester functionalization PP using BPO and that PP interfacial properties can be easily tuned via ester functionalization by SSSP.

**Reactivity of PP-g-ES with Pyr-AA.** Functionalized polyolefins are often used commercially as reactive compatibilizers for immiscible blends.<sup>17,82–85</sup> As further proof of functionalization and to demonstrate reactivity of the PP-g-ES, we investigated the reactivity of the PP-g-ES synthesized by SSSP with Pyr-AA. Under the conditions of reaction, transesterification between PP-g-ES and Pyr-AA is expected, resulting in a covalent attachment of pyrenyl moieties to the PP backbone.<sup>86</sup> The presence of pyrenyl units covalently attached to PP is easily confirmed by fluorescence spectroscopy.<sup>87,88</sup>

Figure 6 compares the fluorescence spectra of a solution of 0.3 g/L Pyr-AA in xylene and a 2.0 g/L PP-g-ES/5 solution in



**Figure 6.** Fluorescence spectra of 0.3 g/L solution of Pyr-MeNH<sub>2</sub> in xylene at 100 °C (dotted curve) and 2 g/L solution PP-g-ES/5 in xylene at 100 °C after first being reacted with Pyr-AA and then purified six times by dissolution and precipitation to remove unreacted Pyr-AA. (Both spectra have had emission intensity normalized to unity at the peak emission wavelength.)

xylene after reaction with Pyr-AA (followed by six cycles of dissolution/precipitation to remove any unreacted Pyr-AA). The spectrum for PP-g-ES/5 shows slight shifts in pyrenyl emission peak wavelengths and structure as compared to that for Pyr-AA; these shifts have been observed in other studies and are associated with modification of photophysical responses (e.g., the polarization, intensity, and energy of the fluorescence transitions) caused by the nature of the chromophore attachment to the polymer backbone.<sup>88–90</sup> Nonetheless, the fluorescence observed for PP-g-ES/5 after reaction with Pyr-AA provides further proof of direct functionalization of PP with benzoates using BPO only and of the potential utility of PP-g-ES as reactive compatibilizer in polymer blends. Control studies carried out on neat PP and PP-g-ES/4MM (synthesized via melt processing), using the same reaction and purification protocol as for PP-g-ES/5, resulted in no fluorescence.

## CONCLUSION

This study demonstrated a novel method of achieving direct ester functionalization of PP using BPO, a symmetric organic peroxide; during SSSP, functionalization is made possible by both thermal and mechanochemical decomposition of BPO at the near-ambient processing  $T$ . At these  $T$  conditions, the extent of decarboxylation of the benzoxyloxy radicals formed during BPO decomposition is significantly decreased, allowing for effective functionalization of PP with ester functional groups (i.e., benzoates), resulting in grafting yields of 36–87%. Contrary to the functionalization realized by SSSP processing, the high- $T$  conditions associated with melt processing result in no grafting for a similar system of PP and BPO because of high extents of decarboxylation. Thus, SSSP provides a new platform of chemistries, which are not attainable via melt processing, for direct functionalization of PP with benzoates.

In addition to achieving PP functionalization with benzoates using BPO, we also demonstrated that this functionalization can be attained with limited reductions in  $M_n$  and  $M_w$ ; for PP-g-ES with 0.18 mol % benzoate groups, only one chain scission event happens for every 12 400 repeat units, resulting in a 17% reduction in  $M_n$  and a 36% reduction in  $M_w$ . These moderate MW reductions are achieved because of the low processing  $T$  employed during SSSP. Under the high  $T$  conditions utilized with melt processing, the extent of  $\beta$ -scission, the primary cause of MW reduction, is high. By processing via SSSP, we strongly suppress the extent of  $\beta$ -scission and thus suppress MW reduction that could accompany functionalization.

Using contact angle measurements and based on a transesterification reaction with Pyr-AA, we provided further proof of successful polar functionalization of PP. We showed that the contact angle of a droplet of deionized water decreased with grafting degree, proving an increase in polarity for PP-g-ES samples relative to the neat, parent PP. Transesterification of PP-g-ES with a fluorescent chromophore was used to demonstrate potential in reactive compatibilization. Finally, the little to no reduction in crystallinity coupled with moderate MW reductions resulted in PP-g-ES samples with little to no losses in mechanical and physical properties relative to the neat parent PP. Together, these results confirm that SSSP can be used to achieve direct ester functionalization of PP to high grafting yields and without the use of a polar monomer.<sup>29,91</sup>

## ASSOCIATED CONTENT

### Supporting Information

Additional rheology data demonstrating the absence of zero shear rate regimes of some PP-g-ES samples, an indication of branching, and high- $T$  GPC chromatograms of some PP-g-ES samples. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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### Notes

The authors declare no competing financial interest.

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